



CS 506: Cognitive Modeling

Course Code	CS 506
Course Name	Cognitive Modeling
L-T-P-C	2-0-2-3
Prerequisites	None
Intended for	3rd, 4th year B. Tech. CSE/M.S. SCEE/Ph.D. SCEE, HSS
Distribution	Elective
Semester	Odd or Even
Approval	3rd Ad-hoc Senate (Nov 2012)

Course Contents

- **Introduction to Cognitive Modeling:** What are cognitive models? Advantages of cognitive models, Practical uses of cognitive models, The steps involved in cognitive modeling
- **Lab:** Install and becoming familiar with cognitive modeling software tools (Excel and/or Matlab) on your computer
- **Qualitative Model Comparisons:** Category learning experiment, Two models of category learning, Qualitative comparisons of Models
- **Lab:** Simulate qualitative model comparison using the Exemplar model in Matlab or Excel
- **Basic Parameter Estimation Techniques:** Linear and Nonlinear parameter estimation, Retention Experiment and Model, Aggregate modeling versus individual modeling, Objective function and searching for optimal parameters
- Application to Choice and Response Time Measures (Signal detection task; Dynamic signal detection model; parameter estimation; goodness of fit; lack of fit tests)
- **Lab:** Simulate parameter estimation using the Retention model in Matlab or Excel. Also, simulate parameter estimation using the Wiener Diffusion Model in Matlab or Excel
- **Quantitative model comparisons:** Maximum likelihood estimation, Becharas Simulated Gambling Task (BSGT), Three Cognitive Models on BSGT, Parameter estimation, Quantitative model comparisons using AIC and BIC, Cross-validation and Generalization
- **Lab:** Create models on BSGT in Matlab or Excel; Simulate parameter estimation on BSGT in Matlab or Excel
- Connectionist versus Rational Approaches: (Rational) Instance-based Learning (Instances; K- nearest neighbor learning; Case-based reasoning; Similarity; Activation), (Connectionist) Neural Networks(neural networks, Rescorla-wagner/delta rule, Multi-layer feed forward networks, Discuss the relative theoretical merits of either approach
- **Lab:** Create and simulate a cognitive model for each of the connectionist and the rational approaches.

Text Books

1. J. Busemeyer & A. Diederich, **Cognitive Modeling**, Sage Publications, 2009. [BD]
2. S. Farrel & S. Stephan Lewandowsky, **Computational Modeling in Cognition: Principles and Practice**, Sage Publications, 2010. [FL]

References

1. R. Sun, **Cognition and Multi-Agent Interaction**, Cambridge University Press, 2006. [RS]
2. Konar, **Artificial Intelligence and Soft Computing: Behavioral and Cognitive Modeling of the Human Brain**, CRC Press, 1999. [AK]
3. T. Mitchell, **Machine Learning**, McGraw-Hill Science, 1997.[TM]
4. B. Hahn, **Essential Matlab for Engineers and Scientists**, 4th Edition, Academic Press, 2009. [BH]

Articles

1. Certain articles from JSTOR: I will hand out photocopies during the semester.

Software

- Matlab and Palisade Decisions Tools (the instructor will provide the CD for these software tools)